

LUMINOUS INTENSITY SENSOR

1. Product Overview

WIN-SN-LUX-65K-2DO-M is a compact, high-precision Light intensity sensor designed for applications requiring low power consumption and small form factors. It ensures reliable measurements over a wide range. Its RS485 communication interface simplifies integration into small, embedded systems, making it ideal for IOT. The sensor's long-term stability ensures consistent performance in harsh environments, making it a cost-effective solution for long-term deployment. In-built DO can be used for ON/OFF control of any 3rd Party system based on intensity of light.

Threshold of the LUX can be set through Modbus. Exceeding threshold will operate a Digital output using relay.



Fig.1 - Product Image

- Luminance to Digital Converter ranging 1 LUX to 65000 LUX.
- Superior sensor performance.
- Operating range -20°C to 60°C and 0% to 70% RH for humidity
- Fully calibrated and processed digital, RS485 Output
- Wide Input voltage options 12-24V DC / 230V AC (optional)
- Excellent long-term stability

Measuring Parameter	Measuring Range	Operating Temperature
Luminous intensity	1-65000 LUX	-20°C ~ 60°C.

2. Sensor Specifications

- **Wide temperature range:** -40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 100*100*60 mm.
- **Certifications:**



3. Power Supply Connection

DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

4. Configuration Settings

- Communication Speed 9600 – 115200 Kbps (SW Selectable)
- Data Bits 8
- Parity None
- CRC Yes
- Slave ID 1, DIP (SW Selectable)
- Function Code 0X03 (Read Holding Register)

Recommended Cable Electrical Characteristics

- 22 AWG Cable** Shielded and twisted pair should be used.
- Tinned Copper** Recommended
- Nominal Conductor DCR** 14.7 ohm / 1000 ft
- Nominal Capacitance** 11 pf / feet (conductor to conductor)
- High Frequency Non-Insertion Loss** 0.5db / 100ft

6. Modbus RS485 Data Storage register

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	Intensity (Lux)	40001	0x03	RS485	16-bit Unsigned int
2	Lux Range	40003	0x03	RS485	16-bit Unsigned int
3	Read & Write Negative offset value	40004	0x03	RS485	16-bit int
4	Read & Write Positive offset value	40005	0x03	RS485	16-bit int
5	Read & Write Lower lux threshold	40006	0x03	RS485	16-bit int
6	Read & Write Higher lux threshold	40007	0x03	RS485	16-bit int
1	Display Baud Rate (Default: 960)	40011	0x03	RS485	16-bit int
2	Enter New Baud Rate	40012	0x03	RS485	16-bit int
3	Display Slave ID (Default: 1)	40013	0x03	RS485	16-bit int
4	Enter New Slave ID	40014	0x03	RS485	16-bit int
5	Display Parity (Default: None-3 And for Odd-1, Even-2)	40015	0x03	RS485	16-bit int
6	Enter New Parity	40016	0x03	RS485	16-bit int
7	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16-bit int
8	Enter New Start/ Stop Bit	40018	0x03	RS485	16-bit int

7. Modbus Configuration

Here is an example of receiving data on Mb poll software.

- Open Mb-poll software connect using configuration setting (Default baud rate is 9600 and slave id is 1).
- After connecting you will get data as shown below.

Important Note

1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc, enter it like this 960/1920/3840/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40011 baud rate (115200) and 2 at 40013 slave id.

Connection Setup

Connection: Serial Port OK Cancel

Serial Settings

USB-SERIAL CH340 (COM15)

9600 Baud

8 Data bits

None Parity

1 Stop Bit

Advanced...

Mode

☒ RTU ☐ ASCII

Response Timeout: 100 [ms]

Delay Between Polls: 10 [ms]

Remote Modbus Server

IP Address or Node Name: 192.168.0.160

Server Port: 502 Connect Timeout: 1000 [ms] ☒ IPv4 ☐ IPv6

Read/Write Definition

Slave ID: 1 OK Cancel

Function: 03 Read Holding Registers (4x)

Address mode

☒ Dec ☐ Hex

Address: 0 PLC address = 40001

Quantity: 18

Scan Rate: 1000 [ms] Apply

Modbus Poll - Mbpoll1

File Edit Connection Setup Functions Display View Window

05 06 15 16 17 22 23

Mbpoll1

Tx = 2978: Err = 405: ID = 1: F = 03: SR = 1000ms

	Name	00000	Name	00010
0	Luxvalue	82	Current Baud	960
1		0	New Baud	0
2	Lux Range	65000	Current SID	1
3	Negative offset	0	New SID	0
4	Positiveoffset	0	Current Parity	3
5	Lower Lux-Relay	0	New Parity	0
6	Higher Lux-Relay	100	Current Sopbit	1
7		0	New Stopbit	0
8		0		
9		0		

- Above image Shows sensor data over Modbus with default Modbus settings
- At the register 40001 shows lux intensity.
- As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.
- To change the baud rate and slave id Jumper should be in single/removed then, you must enter a value on the 40012 resistors for the baud rate and 40014 for the slave id as shown in the image below and restart the board.
- Now you can see data below image set 115200 baud rate and slave ID 10.

Modbus Poll - Mbpoll1

File Edit Connection Setup Functions Display View Window

05 06 15 16 17 22 23

Mbpoll1

Tx = 10: Err = 1: ID = 1: F = 03: SR = 1000ms

	Name	00000	Name	00010
0	Luxvalue	74	Current Baud	960
1		0	New Baud	11520
2	Lux Range	65000	Current SID	1
3	Negative offset	0	New SID	10
4	Positiveoffset	0	Current Parity	3
5	Lower Lux-Relay	0	New Parity	0
6	Higher Lux-Relay	0	Current Sopbit	1
7		0	New Stopbit	0
8		0		
9		0		

Modbus Poll - Mbpoll1

File Edit Connection Setup Functions Display View Window

05 06 15 16 17 22 23

Mbpoll1

Tx = 9: Err = 1: ID = 10: F = 03: SR = 1000ms

	Name	00000	Name	00010
0	Luxvalue	67	Current Baud	11520
1		0	New Baud	0
2	Lux Range	65000	Current SID	10
3	Negative offset	0	New SID	0
4	Positiveoffset	0	Current Parity	3
5	Lower Lux-Relay	0	New Parity	0
6	Higher Lux-Relay	100	Current Sopbit	1
7		0	New Stopbit	0
8		0		
9		0		

- To set the maximum lux range, enter the desired maximum lux value in register 40003. The new value is applied immediately.
- To set the lux negative offset, enter the desired negative offset value in register 40004. The new value is applied immediately, and the updated lux value is reflected in register 40001.
- To set the lux positive offset, enter the desired positive offset value in register 40005. The new value is applied immediately, and the updated lux value is reflected in register 40001.
- Set the lower lux threshold at register 40006. Digital Output 1 is activated when the measured lux value is lower than the threshold value.
- Set the higher lux threshold at register 40007. Digital Output 2 is activated when the measured lux value is higher than the threshold value.

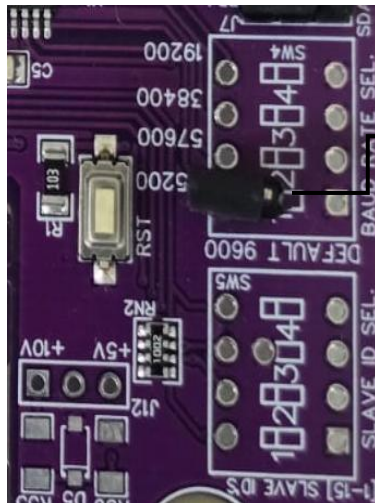


Image1.1



Image1.2

Hard reset Setting

- Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the image 1.2) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

Contact Information

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